Enterprise CRM-to-ERP Agentic Hub

Salesforce CRM & MS Dynamics 365 Business Central Integration Blueprint

1. Project Architecture Blueprint

This specification delivers a comprehensive architectural blueprint for an asynchronous, multi-agent integration engine bridging Salesforce CRM and Microsoft Dynamics 365 Business Central (ERP). The system automates support ticket handling, financial synchronization, and intelligent issue triage while implementing strict deterministic escalation frameworks to isolate complex engineering cases.

2. System Integration Flow Diagram

```
[Salesforce Event Trigger] ──► [FastAPI Webhook Gateway Layer]
                                            │
                                            ▼
                                ┌────────────────────────┐
                                │   Triage & Intent AI   │
                                └────────────────────────┘
                                     │             │
          ┌──────────────────────────┘             └─────────────────────────┐
          ▼ (Intent: Billing/Invoices)                                       ▼ (Intent:
Complex Bugs)
    ┌───────────────────────────────┐
┌──────────────────────────────┐
    │    ERP Sync & Lookup Agent    │                               │  Automated Escalation
Matrix │
    │  (Queries Business Central)   │                               │   (Route to
Engineering)    │
    └───────────────────────────────┘
└──────────────────────────────┘
          │                                                                  │
          ▼                                                                  ▼
[Post Comment & Auto-Resolve Case]                               [Dispatch Slack Alert /
Human HITL]
```

3. Core Technical Implementations

3.1. Webhook Intake Gateway (main.py)

```
from fastapi import FastAPI, BackgroundTasks, Depends, HTTPException
from pydantic import BaseModel, Field
```

```
import uuid

app = FastAPI(title="CRM-to-ERP Agentic Hub Gateway", version="1.0.0")

class SalesforceWebhookPayload(BaseModel):
    case_id: str = Field(..., description="Salesforce Core Event Case Identity Token.")
    subject: str
    description: str
    customer_email: str

async def run_hub_orchestration(payload: SalesforceWebhookPayload):
    # Triggers asynchronous agent evaluation loops
    print(f"[Hub Telemetry] Analyzing Salesforce Case ID: {payload.case_id}")
    pass

@app.post("/webhooks/v1/salesforce-cases", status_code=202)
async def handle_salesforce_case(payload: SalesforceWebhookPayload, background_tasks:
BackgroundTasks):
    background_tasks.add_task(run_hub_orchestration, payload)
    return {"status": "Queued", "message": "Asynchronous multi-agent execution pipeline
triggered."}
```

3.2. Structural Classification Guardrails (classifier.py)

```
from pydantic import BaseModel, Field
from typing import Literal

class TicketClassification(BaseModel):
    intent: Literal["billing_inquiry", "technical_bug", "account_management"] = Field(
        ..., description="The semantic classification derived from the ticket payload
context."
    )
    confidence_score: float = Field(..., description="Statistical boundary probability
score.")
    requires_human_escalation: bool = Field(
        ..., description="Flag determining whether a deep technical engineer must take
full owner authority."
    )
    suggested_internal_notes: str = Field(..., description="Automated logging text
tracking categorization logic.")
```

4. Deployment Matrix & Operational Metrics

Component Target	Technical Domain Responsibility	Enterprise SLA Metric
FastAPI Ingestion Layer	Non-blocking webhook containment and payload parsing	Response Latency < 50ms
Triage & Intent Engine	Zero-shot customer classification and deterministic isolation	Classification Accuracy > 97.8%
Business Central Sync Module	Secure REST querying, ledger validation, and ledger updates	Zero Ledger Discrepancies
Escalation Guardrails	Human-in-the-loop fallback instantiation for out-of-bounds cases	100% Fail-safe Capturing Rate